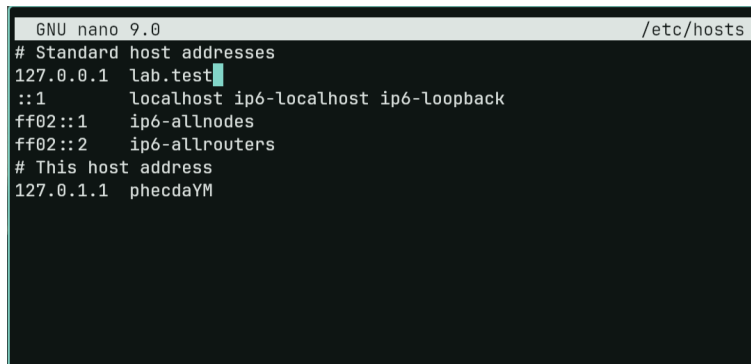


stateful tracking

Younes Menfalouti

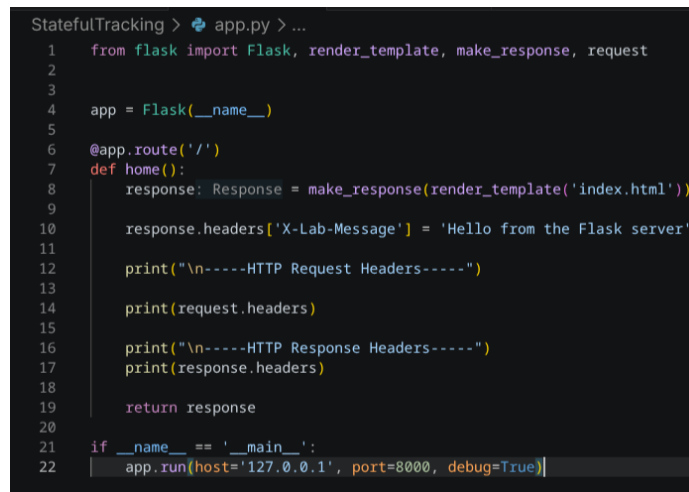
Setup

Every site is a small Flask server on 127.0.0.1. The file `/etc/hosts` maps the lab host names to this address, first `lab.test` and later `publisher1.lab`, `publisher2.lab`, `tracker1.lab` and `analytics.lab`.



```
GNU nano 9.0 /etc/hosts
# Standard host addresses
127.0.0.1 lab.test
::1 localhost ip6-localhost ip6-loopback
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
# This host address
127.0.1.1 phecdam
```

Figure 1. The host name `lab.test` points to 127.0.0.1.



```
StatefulTracking > app.py > ...
1 from flask import Flask, render_template, make_response, request
2
3
4 app = Flask(__name__)
5
6 @app.route('/')
7 def home():
8     response: Response = make_response(render_template('index.html'))
9
10    response.headers['X-Lab-Message'] = 'Hello from the Flask server'
11
12    print("\n-----HTTP Request Headers-----")
13
14    print(request.headers)
15
16    print("\n-----HTTP Response Headers-----")
17    print(response.headers)
18
19    return response
20
21 if __name__ == '__main__':
22     app.run(host='127.0.0.1', port=8000, debug=True)
```

Figure 2. First Flask server with a custom header.

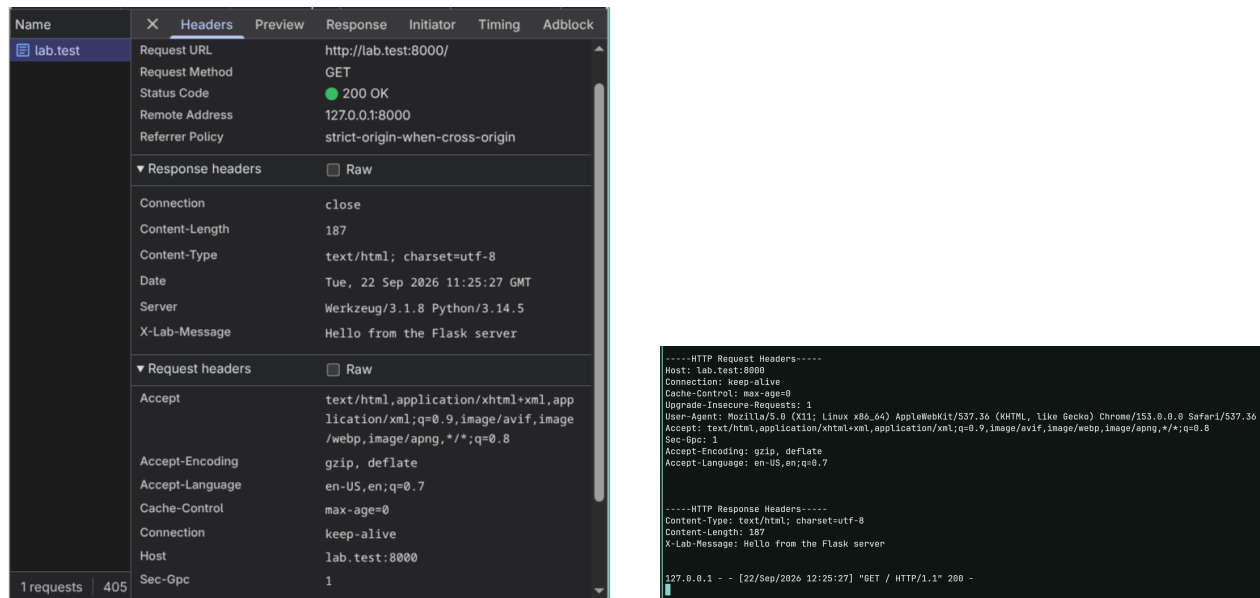


Figure 3. Headers seen in the browser and printed by the server.

Server cookie

The server now reads the cookie aid. When it is missing, the server generates a random value.

```
StatefulTracking > app.py > ...
1 from flask import Flask, render_template, make_response, request
2 import secrets
3
4 app = Flask(__name__)
5
6 @app.route('/')
7 def home():
8     aid: str | None = request.cookies.get('aid')
9     is_new: bool = aid is None
10
11     if is_new:
12         aid: str = secrets.token_hex(8)
13
14         response: Response = make_response(render_template('index.html'))
15
16     if is_new:
17         response.set_cookie(key='aid', value=aid)
18
19     return response
20
21
22 if __name__ == '__main__':
23     app.run(host='127.0.0.1', port=8000, debug=True)
```

Figure 4. The server assigns an identifier on the first visit.

The browser sends `aid=a970c4077d20469a` on every later request. After the cookies are cleared, the server issues a new value `4d15c5c69072dadac`.

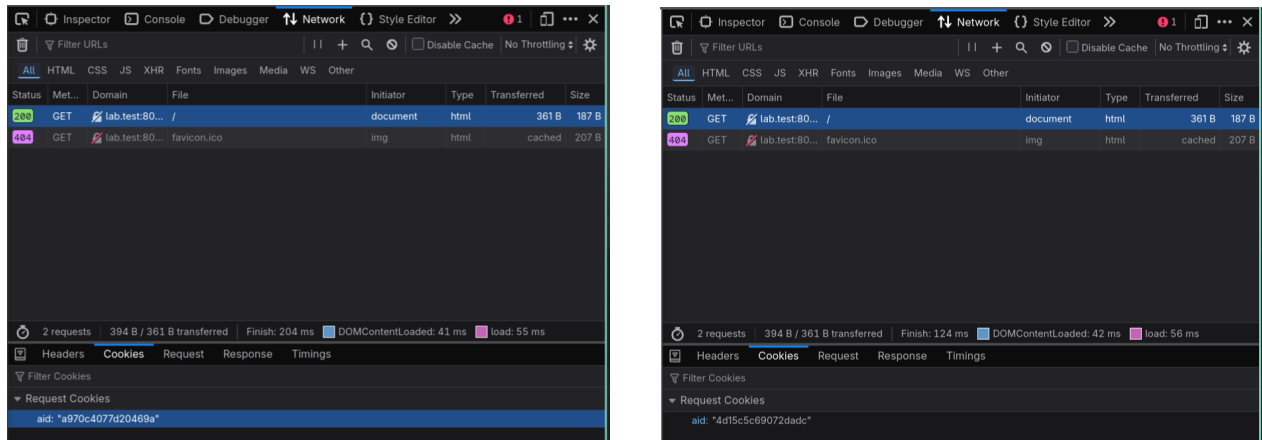


Figure 5. Identifier before and after clearing cookies.

```

--- HTTP REQUEST HEADERS ---
Host: lab.test:8000
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:151.0) Gecko/20100101 Firefox/151.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Accept-Language: en-US,en;q=0.9
Accept-Encoding: gzip, deflate
Connection: keep-alive
Cookie: aid=4d15c5c69072dadcd
Upgrade-Insecure-Requests: 1
Priority: u=0, i

--- HTTP RESPONSE HEADERS ---
Content-Type: text/html; charset=utf-8
Content-Length: 187
X-Lab-Message: Hello from the Flask server

```

Figure 6. The server receives the cookie in the request.

Name	Value	Domain	Path	Expires	Size	HttpOnly	Secure	SameSite	Partition...	Cross Site	Priority
aid	7d01e25f648058ed	lab.test	/	Session	19						Medium

Figure 7. Cookie attributes for lab.test.

Third party tracker

Publisher one (port 8000) shows a technology article and publisher two (port 8001) shows a travel article. Each page embeds an image served by **tracker1.lab** on port 8002. The tracker logs the **aid** cookie, the file and the referer of every image request and sets the cookie on its own domain.

```

StatefulTracking > Tracker-one > app.py > ...
6  app = Flask(__name__)
7
8  IMAGES_DIR = os.path.join(os.path.dirname(__file__), 'images')
9
10 hits: list[Unknown] = []
11
12
13 @app.route('/')
14 def home():
15     aid: str | None = request.cookies.get('aid')
16     is_new: bool = aid is None
17
18     if is_new:
19         aid: str = secrets.token_hex(8)
20
21     response: Response = make_response(render_template('index.html'))
22
23     if is_new:
24         response.set_cookie(key='aid', value=aid)
25
26     response.headers["X-Lab-Message"] = "Hello from the Flask server"
27     print("\n--- HTTP REQUEST HEADERS ---")
28     print(request.headers)
29     print("--- HTTP RESPONSE HEADERS ---")
30     print(response.headers)
31
32     return response
33
34
35 @app.route('/images/<path:filename>')
36 def tracker(filename):
37     aid: str = request.cookies.get('aid') or secrets.token_hex(8)
38     print(f"[TRACKER] aid={aid} file={filename} referer={request.referrer}")
39
40     response: Response = send_from_directory(IMAGES_DIR, filename)
41     response.set_cookie('aid', aid)
42     return response
43
44
45 if __name__ == '__main__':
46     app.run(host='127.0.0.1', port=8002, debug=True)
47

```

Figure 8. Tracker server.

```

StatefulTracking > Publisher-one > templates > index.html > html > body > h1
1  <!DOCTYPE html>
2  <html lang="en">
3
4  <head>
5      <meta charset="UTF-8">
6      <title>Technology Article</title>
7  </head>
8
9  <body>
10     <h1>Welcome to the Technology Article</h1>
11
12     
13
14 </body>
15 </html>
16

```

```

StatefulTracking > Publisher-two > templates > index.html > html > body
1  <!DOCTYPE html>
2  <html lang="en">
3
4  <head>
5      <meta charset="UTF-8">
6      <title>Travel Article</title>
7  </head>
8
9  <body>
10     <h1>Welcome to the Travel Article</h1>
11
12     
13
14 </body>
15 </html>
16

```

Figure 9. Both publishers embed a tracker image.

```

/./
:SSSSS-
:SSSSSSSSS-
/SSSSSSSSSSS+
17-1360P (16) @ 5.00 GHz
-+SSSSSSSSSSSSS+
//SSSSSSSSSSSSS+
`+SSSSSSSSSSSSS+`
+++++

phcda
os EndeavourOS x86_64
cpu 13th Gen Intel(R) Core(TM)
ram 9.07 GiB / 15.28 GiB (59%)
sh zsh 5.9

→ ~ cd VsCode
Fund_ML Info_sec
* VsCode cd Info_sec
StatefulTracking TP_clickjacking
→ Info_sec cd StatefulTracking
Alice-Web-jrney Publisher-two venv
Publisher-one Tracker-one
→ StatefulTracking source venv/bin/activate
(venv) → StatefulTracking cd Tracker-one
app.py images __pycache__ templates
(venv) → Tracker-one python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a prod
uction deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:8002
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 480-261-442

Alice-Web-jrney Publisher-two venv
Publisher-one Tracker-one
→ StatefulTracking source venv/bin/activate
(venv) → StatefulTracking cd Publisher-one
app.py __pycache__ templates
(venv) → Publisher-one python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a prod
uction deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:8000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 840-514-590

Alice-Web-jrney Publisher-two venv
Publisher-one Tracker-one
→ StatefulTracking source venv/bin/activate
(venv) → StatefulTracking cd Publisher-two
app.py __pycache__ templates
(venv) → Publisher-two python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a prod
uction deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:8001
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 832-295-070

```

Figure 10. Tracker and both publishers running.

Each publisher gives the browser its own first party identifier. The publishers cannot link the visits.

```

Host: publisher1.lab:8000
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:151.0) Gecko/20
100101 Firefox/151.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,
*/*;q=0.8
Accept-Language: en-US,en;q=0.9
Accept-Encoding: gzip, deflate
Connection: keep-alive
Cookie: aid=29ffb58666699c2d
Upgrade-Insecure-Requests: 1
Priority: u=0, i
Pragma: no-cache
Cache-Control: no-cache

--- HTTP RESPONSE HEADERS ---

Host: publisher2.lab:8001
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:151.0) Gecko/20
100101 Firefox/151.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,
*/*;q=0.8
Accept-Language: en-US,en;q=0.9
Accept-Encoding: gzip, deflate
Connection: keep-alive
Cookie: aid=d03b6b84e5209913
Upgrade-Insecure-Requests: 1
Priority: u=0, i
Pragma: no-cache
Cache-Control: no-cache

--- HTTP RESPONSE HEADERS ---

```

Figure 11. Different first party identifiers on the two publishers.

The tracker sees the same identifier on both publishers and on its own site. The browser sends

the `tracker1.lab` cookie with every image request, whatever page embeds the image. The referer tells the tracker which page the user reads, so it builds one profile containing the technology article and the travel article.

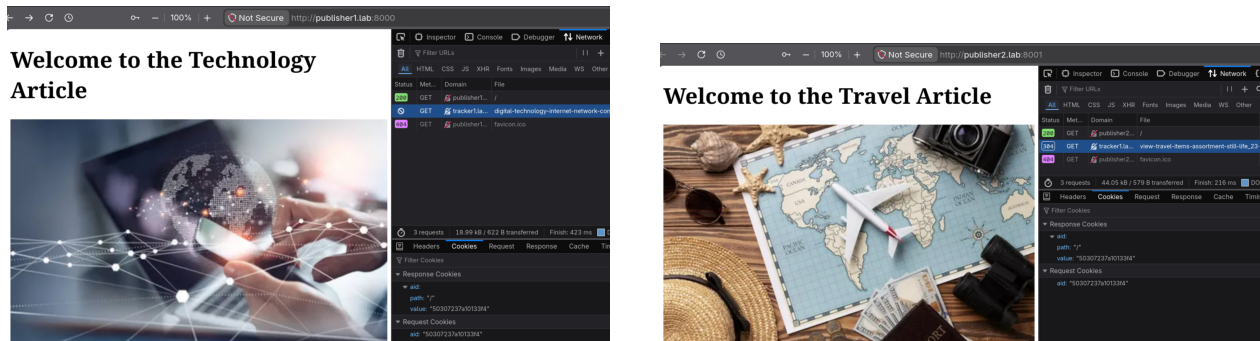


Figure 12. The same tracker cookie on publisher one and publisher two.

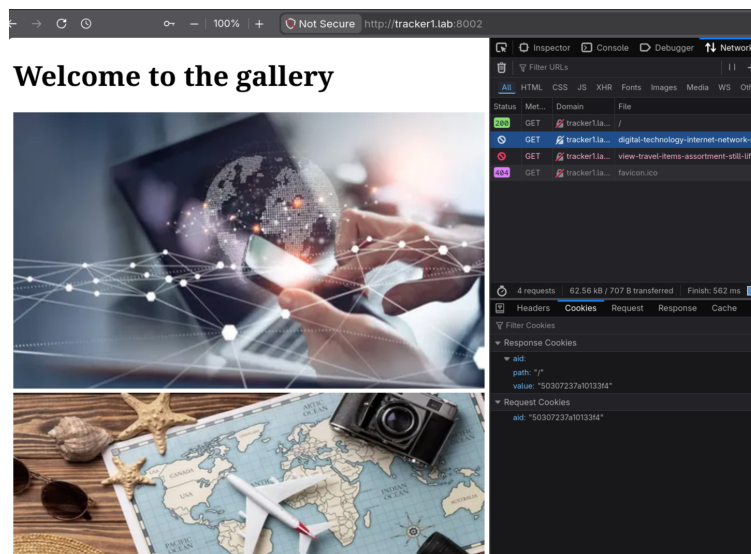


Figure 13. The same tracker cookie on the tracker site.

```

127.0.0.1 - - [25/Sep/2026 21:38:09] "GET / HTTP/1.1" 200 -
[TRACKER] aid=50307237a10133f4 file=digital-technology-interne
t-network-connection-big-data-digital-marketing-iot-internet-o
f.webp referer=http://tracker1.lab:8002/
127.0.0.1 - - [25/Sep/2026 21:38:10] "GET /images/digital-tech
nology-internet-network-connection-big-data-digital-marketing-
iot-internet-of.webp HTTP/1.1" 304 -
[TRACKER] aid=50307237a10133f4 file=view-travel-items-assortme
nt-still-life_23-2149617645.webp referer=http://tracker1.lab:8
002/
127.0.0.1 - - [25/Sep/2026 21:38:10] "GET /images/view-travel-
items-assortment-still-life_23-2149617645.webp HTTP/1.1" 304 -
[TRACKER] aid=50307237a10133f4 file=view-travel-items-assortme
nt-still-life_23-2149617645.webp referer=http://publisher2.lab
:8001/
127.0.0.1 - - [25/Sep/2026 21:38:11] "GET /images/view-travel-
items-assortment-still-life_23-2149617645.webp HTTP/1.1" 304 -
[TRACKER] aid=50307237a10133f4 file=digital-technology-interne
t-network-connection-big-data-digital-marketing-iot-internet-o
f.webp referer=http://publisher1.lab:8000/
127.0.0.1 - - [25/Sep/2026 21:38:13] "GET /images/digital-tech
nology-internet-network-connection-big-data-digital-marketing-
iot-internet-of.webp HTTP/1.1" 304 -
[TRACKER] aid=50307237a10133f4 file=view-travel-items-assortme
nt-still-life_23-2149617645.webp referer=http://publisher2.lab
:8001/
127.0.0.1 - - [25/Sep/2026 21:40:07] "GET /images/view-travel-
items-assortment-still-life_23-2149617645.webp HTTP/1.1" 304 -

```

Figure 14. Tracker log with one identifier across all referers.

First party analytics

The analytics run on `analytics.lab` at port 9100 and serves `analytics.js`. Both publishers load it with a script tag. The script looks for its cookie, creates a random identifier when none exists.

```

StatefulTracking > Analytics-one > app.py > ...
1  from flask import Flask, request
2
3  app = Flask(__name__)
4
5
6  @app.route('/')
7  def home():
8      uid: str | None = request.args.get('uid')
9      page: str | None = request.args.get('page')
10     print(f"[ANALYTICS] uid={uid} page={page} referer={request.referrer} cookies={dict(request.cookies)}")
11     return ''
12
13
14 if __name__ == '__main__':
15     app.run(host='127.0.0.1', port=9100, debug=True)
16

```

Figure 15. Analytics server.

```

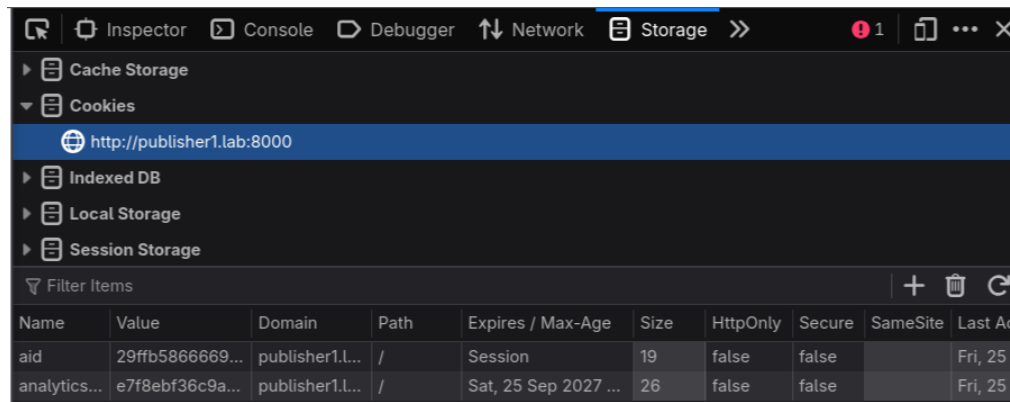
StatefulTracking > Analytics-one > static > JS analytics.js > ...
1  var uid = null;
2  var cookies = document.cookie.split('; ');
3
4  for (var i = 0; i < cookies.length; i++) {
5      if (cookies[i].startsWith('analytics_id=')) {
6          uid = cookies[i].split('=')[1];
7      }
8  }
9
10 if (uid === null) {
11     uid = Math.random().toString(16).slice(2);
12     document.cookie = 'analytics_id=' + uid + '; max-age=31536000; path=/';
13 }
14
15 new Image().src = 'http://analytics.lab:9100/?uid=' + uid + '&page=' + location.href;
16

```

Figure 16. The script `analytics.js`.

The script comes from `analytics.lab` but runs inside the publisher page, so `document.cookie` writes a cookie for the publisher. The two publishers receive different identifiers.

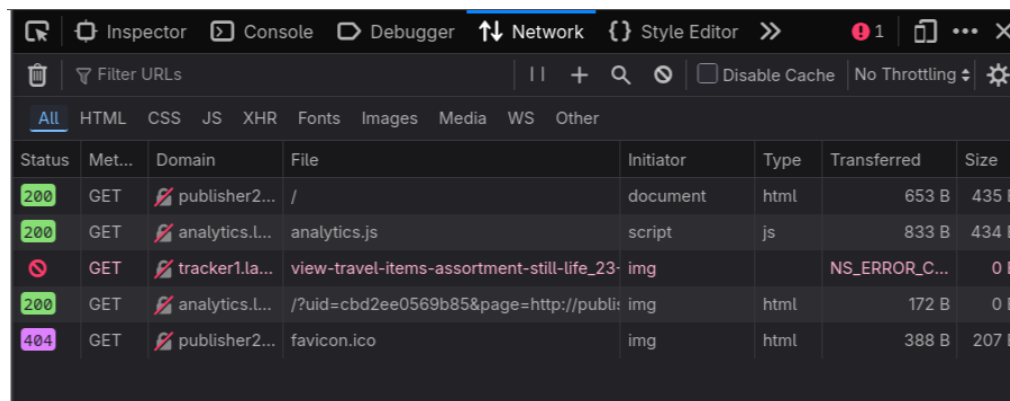
The storage panel of publisher one shows the analytics cookie



Name	Value	Domain	Path	Expires / Max-Age	Size	HttpOnly	Secure	SameSite	Last Ac
aid	29ffb5866669...	publisher1.L...	/	Session	19	false	false		Fri, 25
analytics...	e7f8ebf36c9a...	publisher1.L...	/	Sat, 25 Sep 2027 ...	26	false	false		Fri, 25

Figure 17. Cookies stored for publisher one.

In the network panel of publisher two, the page loads `analytics.js` from `analytics.lab` as a script.



Status	Met...	Domain	File	Initiator	Type	Transferred	Size
200	GET	publisher2...	/	document	html	653 B	435 B
200	GET	analytics.L...	analytics.js	script	js	833 B	434 B
200	GET	tracker1.la...	view-travel-items-assortment-still-life_23-	img	NS_ERROR_C...	0 B	0 B
200	GET	analytics.L...	?uid=cbd2ee0569b85&page=http://publi...	img	html	172 B	0 B
404	GET	publisher2...	favicon.ico	img	html	388 B	207 B

Figure 18. Analytics requests on publisher two.

The analytics server does not receive these cookies automatically. The browser only attaches

cookies of `analytics.lab` to requests for `analytics.lab`, and that domain has none, so the server always prints an empty cookie list. It learns the identifier only because the script copies it into the query string.

```
[ANALYTICS] uid=None page=None referer=None cookies={}
127.0.0.1 - - [25/Sep/2026 22:13:38] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [25/Sep/2026 22:13:38] "GET /favicon.ico HTTP/1.1" 404 -
[ANALYTICS] uid=None page=None referer=None cookies={}
127.0.0.1 - - [25/Sep/2026 22:13:42] "GET / HTTP/1.1" 200 -
* Detected change in '/home/phedda/VsCode/Info_sec/StatefulTracking/Analytics-one/app.py', reloading
* Restarting with stat
* Debugger is active!
* Debugger PIN: 480-261-442
[ANALYTICS] uid=None page=None referer=None cookies={}
127.0.0.1 - - [25/Sep/2026 22:16:56] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [25/Sep/2026 22:17:40] "GET /static/analytics.js HTTP/1.1" 200 -
127.0.0.1 - - [25/Sep/2026 22:18:34] "GET /static/analytics.js HTTP/1.1" 200 -
[ANALYTICS] uid=e7f8ebf36c9a98 page=http://publisher1.lab:8000/ referer=http://publisher1.lab:8000/ cookies={}
127.0.0.1 - - [25/Sep/2026 22:18:34] "GET /?uid=e7f8ebf36c9a98&page=http://publisher1.lab:8000/ HTTP/1.1" 200 -
127.0.0.1 - - [25/Sep/2026 22:18:38] "GET /static/analytics.js HTTP/1.1" 200 -
[ANALYTICS] uid=cdb2ee0569b85 page=http://publisher2.lab:8001/ referer=http://publisher2.lab:8001/ cookies={}
127.0.0.1 - - [25/Sep/2026 22:18:38] "GET /?uid=cdb2ee0569b85&page=http://publisher2.lab:8001/ HTTP/1.1" 200 -
[ANALYTICS] uid=None page=None referer=None cookies={}
127.0.0.1 - - [25/Sep/2026 22:18:40] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [25/Sep/2026 22:18:40] "GET /favicon.ico HTTP/1.1" 404 -
```

Figure 19. Requests received by the analytics server.

The log shows two profiles. The lines with no identifier come from opening the analytics server directly. The service builds separate profiles because a first party cookie never leaves its publisher. The third party tracker merges both visits into one profile.